Worksheet - Electric force and electric field

Understanding the electric force:

- If we have one positive charge and one negative charge in a box, do the charges attract (move towards each other) or repel (move away from each other)?

Solution: These charges are opposites of one-another so they will attract (move towards each other).

- If we have two positive charges in a box, do the charges attract (move towards each other) or repel (move away from each other)?

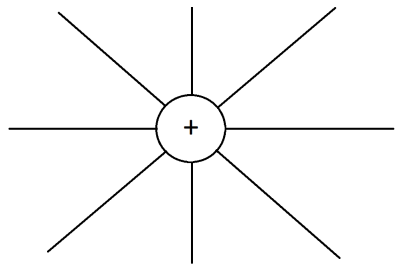
Solution: These charges are the same as one-another so they will repel (move away from each other).

- If we have two negative charges in a box, do the charges attract (move towards each other) or repel (move away from each other)?

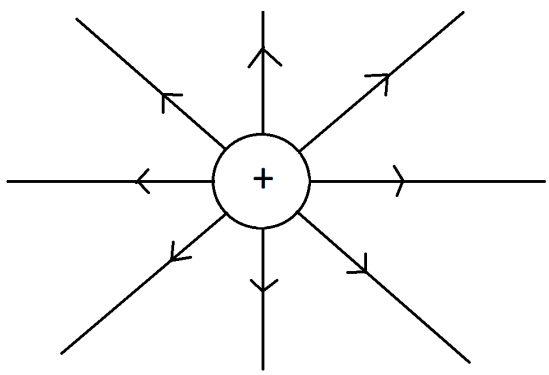
Solution: These charges are the same as one-another so they will repel (move away from each other).

Understanding the electric fields of one charge:

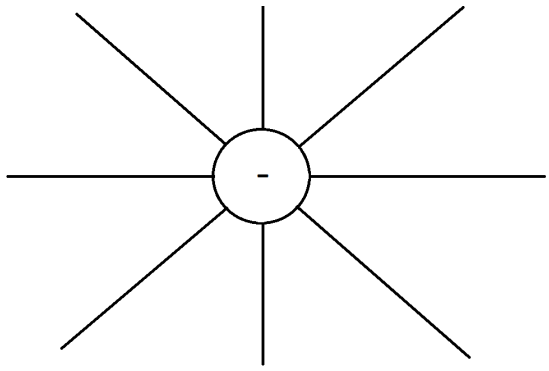
- In this picture we have one positive charge with the field lines drawn. I have not drawn the arrows. Do the arrows point towards the circle or away from the circle?



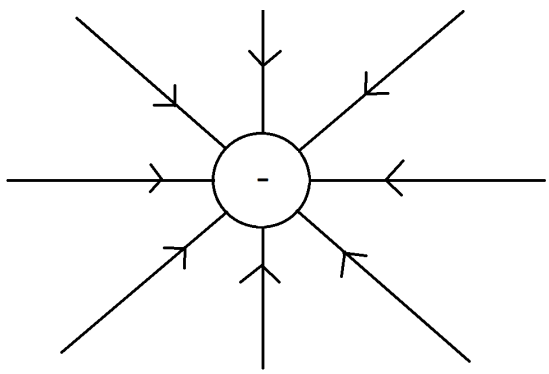
Solution: The charge is positive so the arrows will point away from the circle (the circle is the charge here).



- In this picture we have one negative charge with the field lines drawn. I have not drawn the arrows. Do the arrows point towards the circle or away from the circle?

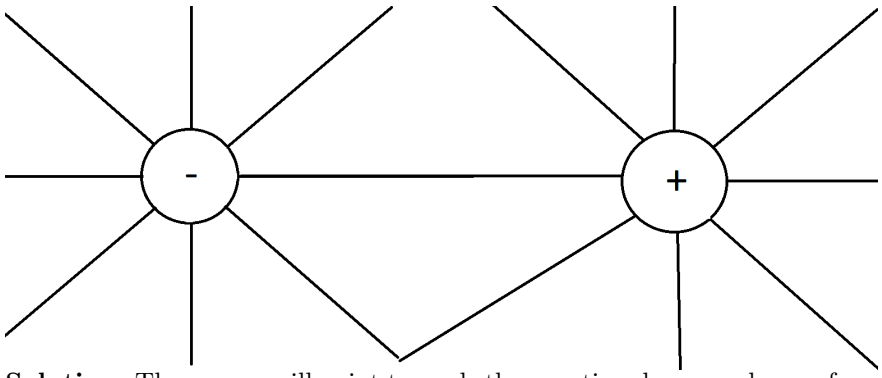


Solution: The charge is negative so the arrows will point towards the circle (the circle is the charge here).

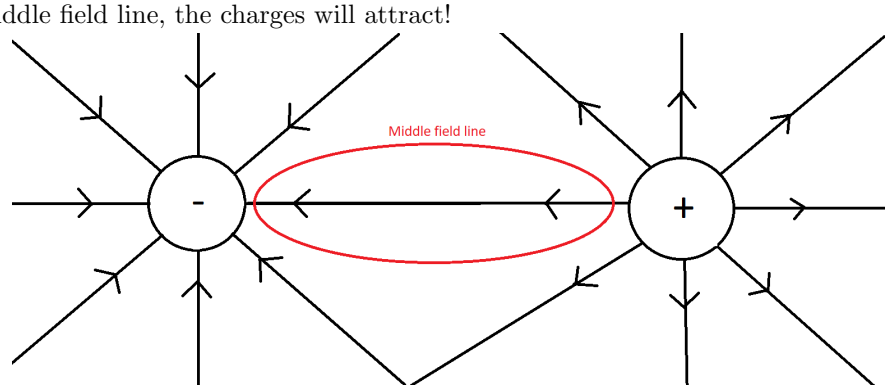


Understanding the electric fields of two charges:

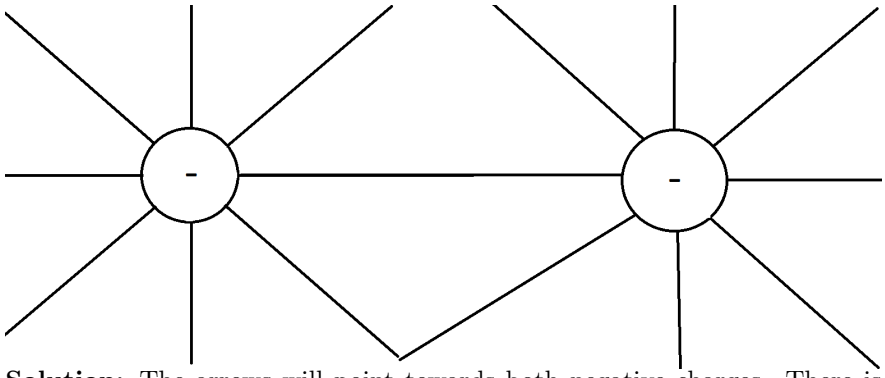
- In this picture we have one negative charge and one positive charge with the field lines drawn. I have not drawn the arrows. Do the arrows point towards the negative charge or away from it? Do the arrows point toward the positive charge or away from it? Do you expect them to attract or repel?



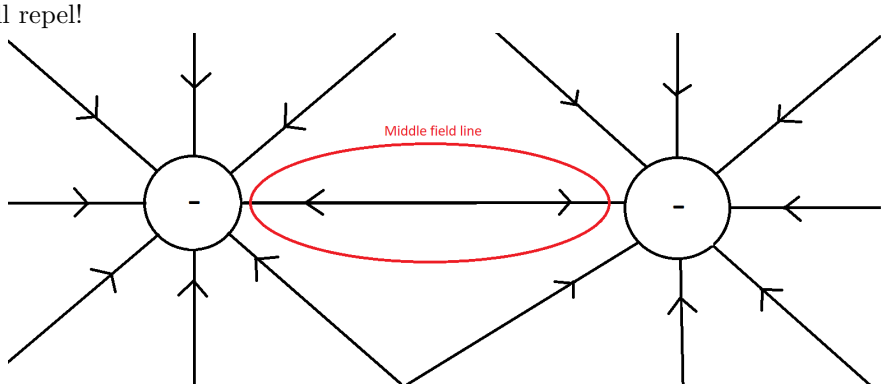
Solution: The arrows will point towards the negative charge and away from the positive charge. There is a field line that connects the negative and positive charge in the middle. Since the arrows point in the same direction along this middle field line, the charges will attract!



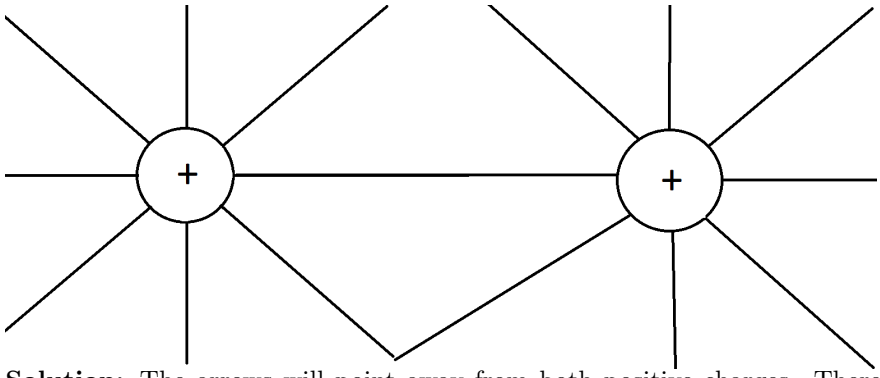
- In this picture we have two negative charges with the field lines drawn. I have not drawn the arrows. Do the arrows point towards the negative charge or away from it? Do the arrows point toward the positive charge or away from it? Do you expect them to attract or repel?



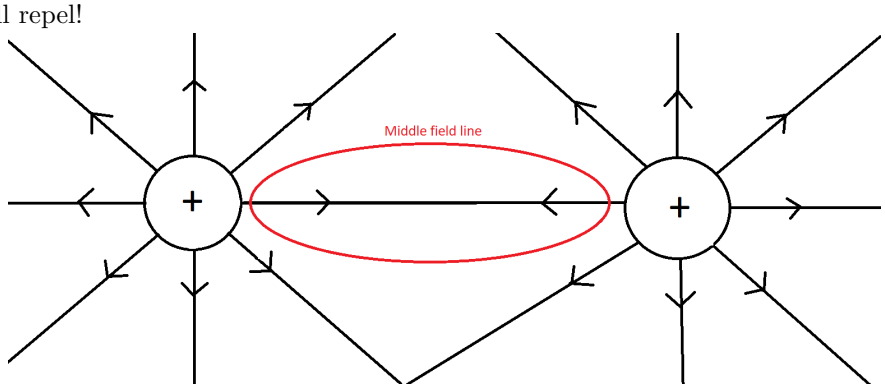
Solution: The arrows will point towards both negative charges. There is a field line that connects the two negative charges in the middle. Since the arrows point in the opposite directions along this middle field line, the charges will repel!



- In this picture we have two positive charges with the field lines drawn. I have not drawn the arrows. Do the arrows point towards the negative charge or away from it? Do the arrows point toward the positive charge or away from it? Do you expect them to attract or repel?

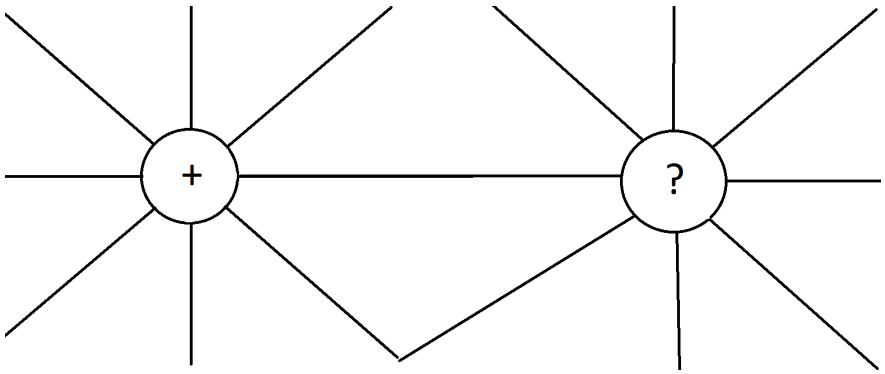


Solution: The arrows will point away from both positive charges. There is a field line that connects the two positive charges in the middle. Since the arrows point in the opposite directions along this middle field line, the charges will repel!

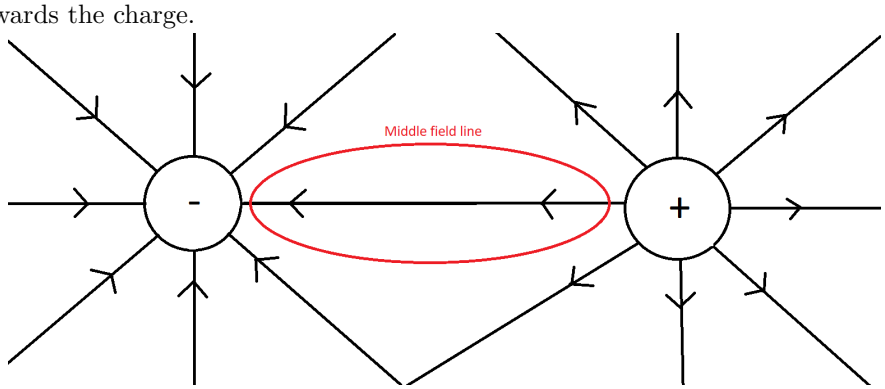


Mystery charge:

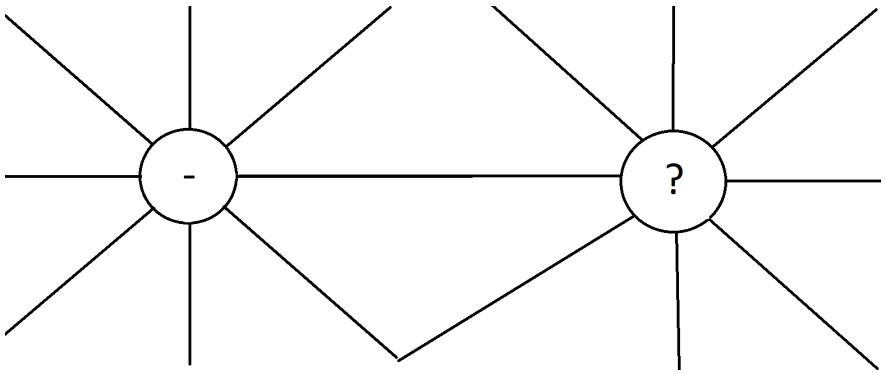
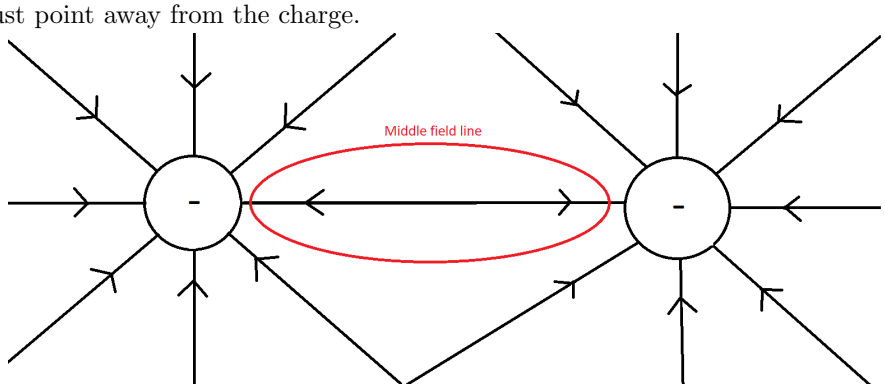
- In this picture we have a positive charge and a mystery charge with the field lines drawn. I have not drawn the arrows. The charges attract. Is the mystery charge positive or negative? Do the arrows point towards the mystery charge or away from it? Do the arrows point toward the positive charge or away from it?



Solution: If the charges attract, they must be opposite charges. One charge is positive so the mystery charge must be negative. The arrows point away from the positive charge. Since the mystery charge is negative, the arrows must point towards the charge.



- In this picture we have a negative charge and a mystery charge with the field lines drawn. I have not drawn the arrows. The charges repel. Is the mystery charge positive or negative? Do the arrows point towards the mystery charge or away from it? Do the arrows point toward the negative charge or away from it?



Solution: If the charges repel, they must both be the same charge. One charge is positive so the mystery charge must be positive. The arrows point away from the positive charge. Since the mystery charge is positive, the arrows must point away from the charge.